

Solar Energy Production & Revenue Forecasting Dashboard

Phoenix, USA vs. London, UK — a comparative analysis of solar viability, revenue, and investment return across two markets

Data Sources: NASA POWER API · U.S. EIA API · Ofgem Energy Price Cap

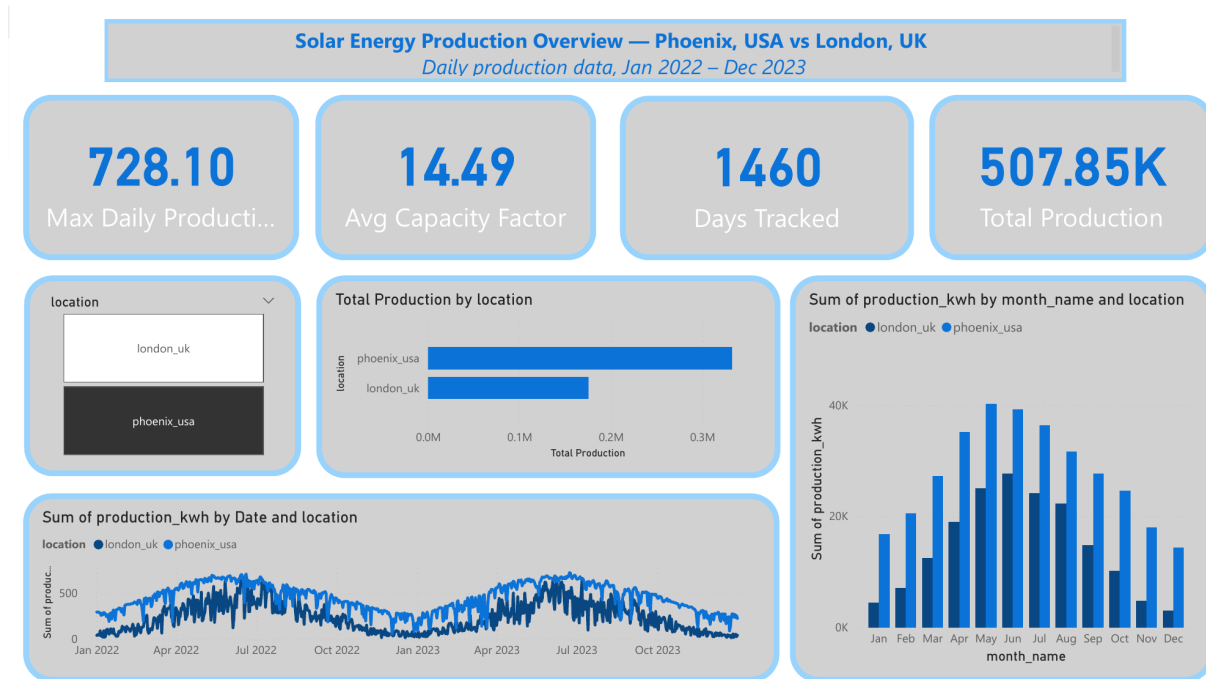
Tools: Python (data pipeline) · Power BI (dashboard & DAX modeling)

Period Covered: January 2022 – December 2023 (historical) | January – March 2024 (forecast)

Headline finding: Phoenix produces nearly double the solar energy of London, yet London achieves a faster investment payback and a higher 25-year ROI — driven by substantially higher UK electricity prices. The analysis shows that revenue potential depends as much on local market pricing as it does on raw solar resource availability.

Production Overview

Daily production data, Jan 2022 – Dec 2023

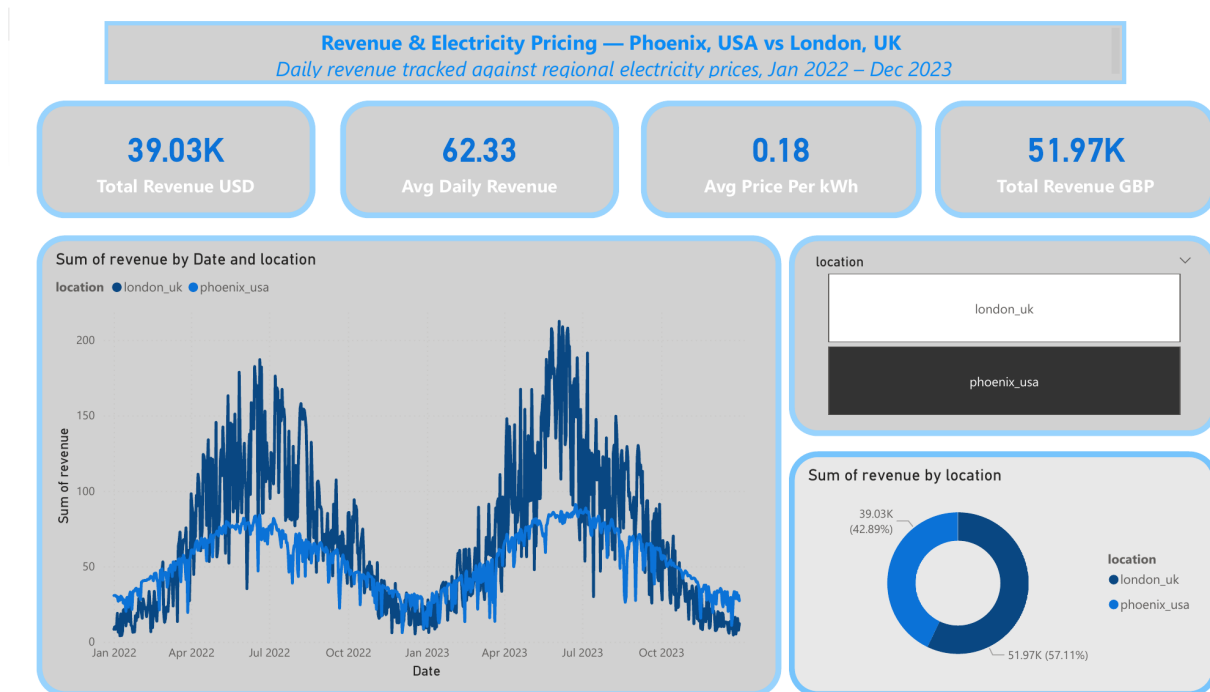


This page tracks daily solar output for two hypothetical 100kW installations over a two-year period. Phoenix delivers consistently high, stable production year-round, supported by reliable desert sun exposure and an average capacity factor well above London's. London, by contrast, shows a pronounced seasonal swing — strong output through the summer months collapsing sharply each winter.

Insight: Total production confirms Phoenix generates significantly more raw energy overall (507.85K kWh combined across both sites, with Phoenix contributing the larger share) — setting up the counterintuitive revenue comparison that follows on Page 2.

Revenue & Market Price

Daily revenue tracked against regional electricity prices, Jan 2022 – Dec 2023

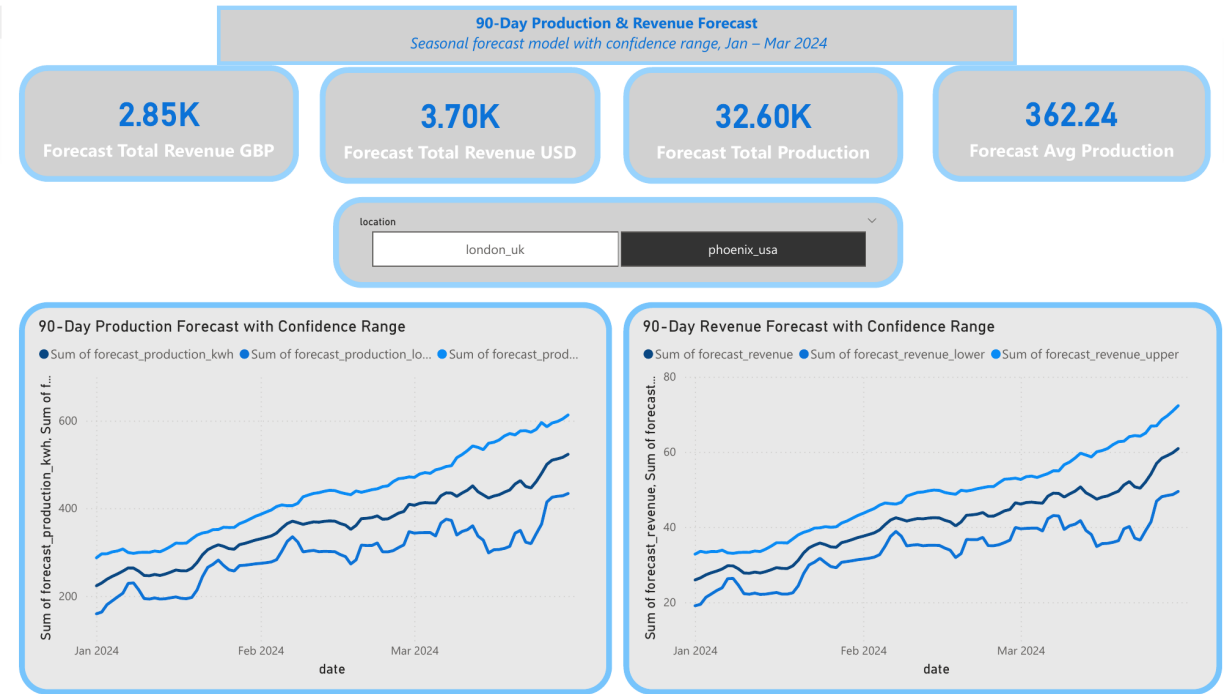


Despite producing roughly half the energy of Phoenix, London generates more total revenue (\$39.03K USD for Phoenix vs. £51.97K GBP for London) — a direct result of substantially higher UK electricity prices, particularly during the 2022–2023 European energy crisis. Revenue figures for each market are kept in their native currency throughout, since blending USD and GBP would distort the comparison rather than clarify it.

Insight: Electricity pricing, not production volume, is the dominant driver of revenue outcomes in this analysis — a finding with direct implications for where solar investment capital is best deployed.

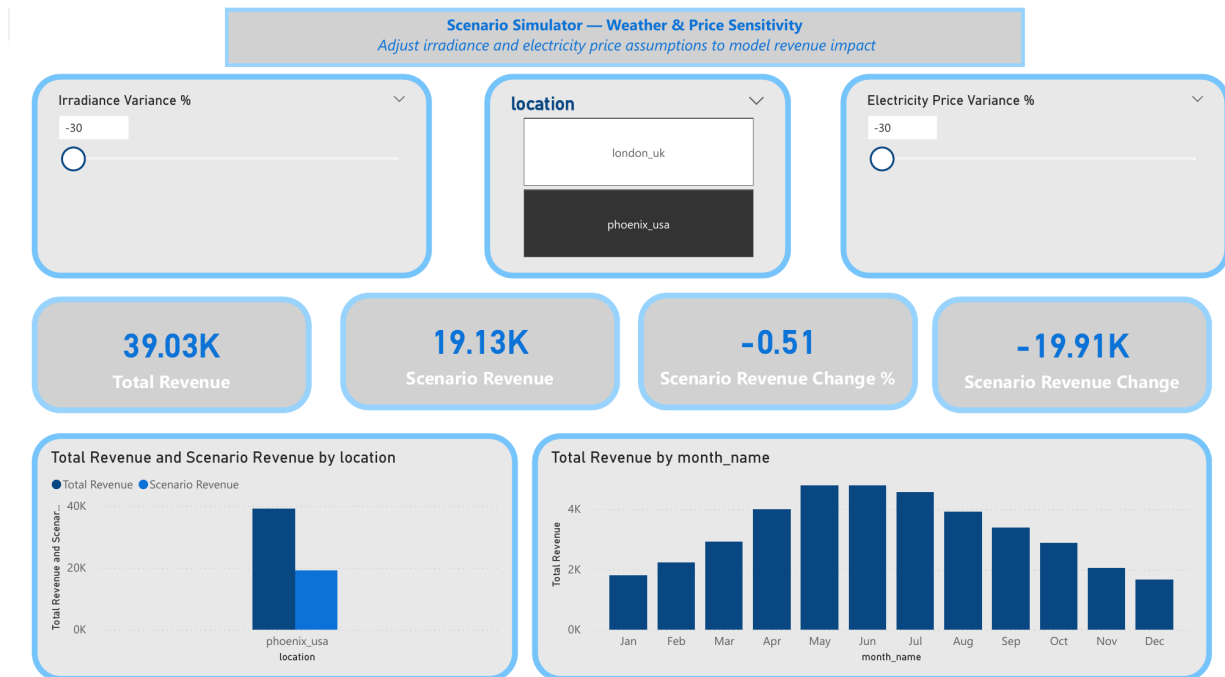
Forecast

Seasonal forecast model with confidence range, Jan – Mar 2024



Scenario Simulator

Adjust irradiance and electricity price assumptions to model revenue impact



An interactive what-if tool built with Power BI parameters, letting a user test how changes in solar irradiance and electricity pricing — independently or together — affect projected revenue. Useful for stress-testing investment assumptions before committing capital, or for understanding downside risk under less favorable weather or market conditions.

Insight: A simultaneous 30% drop in both irradiance and electricity price compounds to roughly a 51% revenue decline — demonstrating that these two risk factors interact multiplicatively rather than additively, an important consideration for risk modeling.

ROI & Break-Even Analysis

Investment payback and lifetime return for a 100kW solar installation



Closes the analysis with the investment decision itself. Using realistic commercial installed-cost assumptions for each market, the model calculates simple payback period, break-even electricity price, and 25-year lifetime ROI based on actual historical production and revenue performance.

Insight: Both locations achieve payback within six years and exceed 340% lifetime ROI. London's higher electricity prices result in a faster payback (5.39 years) and higher overall return (364% lifetime ROI) than Phoenix (5.64 years, 343.5% ROI) — despite producing significantly less solar energy. This is the analysis's central finding: revenue potential depends as much on market pricing as on resource availability.